



Enterprise AI & jMetal UCDP Solver (Modular)

Uncapacitated Facility Location Problem for Business Optimization

Architecture: Separation of Concerns (Frontend, jMetal Backend, AI Analyzer Backend)



Business Setup

Potential Warehouse Locations:

A (Alvalade), B (Benfica), C (Cascais)

Number of Active Customers:



Run Hybrid Optimization



Dashboard Executive Summary

Total Optimized Cost (jMetal)

€18,994

Fixed Setup Cost

€12,000

Transportation Cost

€6,994

Recommended Locations to Open: B (Benfica)



Business Justification

The selection of opening only one facility at 'B (Benfica)' optimizes the total cost by reducing both fixed and variable costs to a total optimized cost of 18994 EUR. This solution serves all 20 customers efficiently, minimizing transportation costs while keeping fixed setup costs manageable. The single location strategy ensures that no customer is left unserved, improving service quality, which aligns with the company's strategic goals.



Risk Assessment

One potential risk associated with this setup is operational resilience in case of unforeseen disruptions at 'B (Benfica)'. This could lead to an interruption in supply chain services for some customers if that specific location becomes unavailable or suffers a significant downtime.



Management Recommendations

To mitigate the identified risk, it's recommended that management should diversify their logistics by establishing backup suppliers or facilities closer to other major customer bases. This can enhance operational resilience and reduce potential service disruptions.

View Raw Backend JSON Data

```
{
  "business_justification": "The selection of opening only one facility at 'B (Benfica)' optimizes the total cost by reducing both fixed and variable costs to a total optimized cost of 18994 EUR. This solution serves all 20 customers efficiently, minimizing transportation costs while keeping fixed setup costs manageable. The single location strategy ensures that no customer is left unserved, improving service quality, which aligns with the company's strategic goals.",
  "risk_assessment": "One potential risk associated with this setup is operational resilience in case of unforeseen disruptions at 'B (Benfica)'. This could lead to an interruption in supply chain services for some customers if that specific location becomes unavailable or suffers a significant downtime.",
  "recommendations": "To mitigate the identified risk, it's recommended that management should diversify their logistics by establishing backup suppliers or facilities closer to other major customer bases. This can enhance operational resilience and reduce potential service disruptions."
}
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